Integration: Surface Area

Questions 1–20

Question 1. In each case, the x-coordinate of two points P and Q and the curve on which they lie are given. The arc PQ is completely rotated about the x-axis. Determine the exact value of the surface area generated.

a) $x_P = 0, x_Q = 2, y = 3x + 5$
b) $x_P = 0, x_Q = 3, y = 2\sqrt{x}$
c) $x_P = 0, x_Q = 1, y = \cosh x$
d) $x_P = 1, x_Q = 4, y = \frac{1}{2\sqrt{3}}\sqrt{x}(4 - x)$

Question 2. A line is given by the equation $y = ax$ for a positive constant a, between points $x = 0$ and $x = k$ for positive constant k. Use integration to determine the surface area generated if the arc is rotated 2π radians about the x-axis.

Question 3. A small decorative bowl is modelled by

$$y^2 = x + 16$$

for $0 \leq x \leq 6$, where x and y are measured in centimetres. The section of the curve between $x = 0$ and $x = 6$ is rotated completely about the x-axis to form the curved surface of the bowl.

a) Find the surface area of the bowl giving your answer to four significant figures.

b) The cost of one coating of paint for the bowl is £0.03 per cm^2. Find the cost of coating 20 bowls to the nearest £1

Question 4. A curve is given by equation $y = \sqrt{x+1}$
An arc on the curve lies between the points where $x = 0$ and $x = k$ and is rotated completely about the x-axis. The surface area generated is given by $\frac{\pi}{6}(13\sqrt{13} - 5\sqrt{5})$ units^2. Determine the value of k

Question 5. A curve is given by equation $y = \sqrt{k - x^2}$ for positive constant k between points P and Q which lie on the x-axis and y-axis respectively. By finding the surface area generated when the arc is rotated 2π radians about the x-axis, find the surface area of a sphere with radius $\sqrt{k}$ giving your answer in terms of k

Question 6.

a) By using the substitution $x = \sinh t$, show that

$$\int \sqrt{1 + x^2}\, dx = \frac{1}{2} x \sqrt{x^2 + 1} + \frac{1}{2} \ln(x + \sqrt{x^2 + 1}) + c$$

b) By using the substitution $u = 2e^{2x}$, show that the area of the surface generated when the curve $y = e^{2x}$, for $0 \leq x \leq \ln 2$, is rotated about the x-axis is 48.2 units^2, correct to three significant figures.

Question 7. A curve is given by equation

$$y = \frac{1}{3}\cos(3x)$$

An arc on the curve lies between the points where $x = 0$ and $x = \frac{\pi}{6}$ is rotated through 2π radians about the x-axis to form surface area S

a) Show that S is given by

$$S = \frac{2\pi}{3}\int_0^{\pi/6} \cos(3x)\sqrt{1+\sin^2(3x)}\,dx$$

b) Use the substitution $u = \sin(3x)$ to find the exact value of S

Question 8. In each case, the x-coordinate of two points P and Q and the curve on which they lie are given. The arc PQ is completely rotated about the y-axis. Determine the exact value of the surface area generated.

a) $x_P = 2, x_Q = 3, y = \frac{\sqrt{7}}{3}(x+1)$
b) $x_P = 1, x_Q = 2, y = x^2$
c) $x_P = 0, x_Q = 5, y = \frac{2}{3}(x+3)^{3/2}$
d) $x_P = 0, x_Q = \ln 2, y = \cosh x$

Question 9. A line is given by the equation $y = ax$ for a positive constant a, between points $x = 0$ and $x = k$ for positive constant k. Use integration to determine the surface area generated if the arc is rotated 2π radians about the y-axis.

Question 10. A curve is given by the equation $y = \operatorname{arcosh} x$
An arc on the curve, s, lies between the points where $x = 1$ and $x = 2$ and is rotated completely about the y-axis. Show that the area of the surface generated is given by $(2\pi\sqrt{3} + \pi \ln(2 + \sqrt{3}))$ units^2

Parametric Curves

Question 11. Find the exact area of the surface generated by rotating the following arcs 2π radians about the x-axis:

a) $x = 6t, y = 2t$ for $0 \leq t \leq 3$

b) $x = t^2, y = \sqrt{4t^2 + 1}$ for $0 \leq t \leq 1$

Question 12. Find the exact area of the surface generated by rotating the following arcs 2π radians about the y-axis:

a) $x = 4t^2, y = t^2$ for $1 \leq t \leq 2$

b) $x = \sin^2 t, y = \cos^2 t$ for $0 \leq t \leq \frac{\pi}{2}$

Question 13. A curve is defined by the parametric equations

$$x = r\cos\theta, y = r\sin\theta$$

where r is a positive constant and $0 \leq \theta \leq \pi$
By rotating the curve about the x-axis, use integration to show that the surface area of a sphere is $4\pi r^2$

Question 14. A curve is defined by the parametric equations

$$x = \operatorname{sech} t, y = \tanh t$$

for $t \geq 0$
The curve is rotated 2π radians about the x-axis between the points $t = 0$ and $t = \ln 5$

a) Show that $\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2 = \operatorname{sech}^2 t$

b) Find the exact area of the surface area generated.

Question 15. The shape of a funnel may be modelled using parametric equations

$$x = 4(1+t)^{3/2}, y = 4(1-t)^{3/2}$$

where x and y are measured in centimetres for $0 \leq t \leq 0.8$ which is rotated 2π radians about the x-axis.

a) Determine the surface area of the funnel giving your answer to three significant figures.

The manufacturer wants to add a single layer of protective coating to the funnels. The cost of one layer of protective coating for the funnel is £0.02 per cm.

b) Find the cost of covering 15 funnels with one layer to the nearest £1

Question 16. A curve is defined by the parametric equations

$$x = k\cos^3 t, y = k\sin^3 t$$

where k is a positive constant r and $\frac{\pi}{4} \leq t \leq \frac{\pi}{2}$

The curve is rotated 2π radians about the x-axis. Use integration to show that the surface area formed is $\frac{3\pi k^2}{20}(8-\sqrt{2})$ units^2.

Question 17. A curve is defined by the parametric equations

$$x = 2t, y = \frac{1}{2}t^2 - \ln t$$

for $1 \leq t \leq 2$
The curve is rotated 2π radians about the y-axis. Determine the exact area of the surface area generated.

Question 18.

a) Use integration by parts to show that

$$\int \frac{\ln x}{x}\, dx = \frac{1}{2}\ln^2 x + c$$

A curve is defined by the parametric equations

$$x = 2t, y = \frac{1}{2}t^2 - \ln t$$

for $1 \leq t \leq 3$. The curve is rotated 2π radians about the x-axis.

b) Find the area of the surface area generated giving your answer to three significant figures.

Question 19. A curve is defined by the parametric equations

$$x = e^{\theta}\sin\theta, y = e^{\theta}\cos\theta$$

for $0 \leq \theta \leq \frac{\pi}{2}$. The curve is rotated 2π radians about the x-axis. Use integration to show that the surface area formed is $\frac{2\pi\sqrt{2}}{5}(e^{\pi} - 2)$ units^2.

Question 20.

a) Show that $\int e^{-2x}(\cos x - \sin x)\,dx = -\frac{1}{5}e^{-2x}(\cos x - 3\sin x) + c$

A curve is defined by the parametric equations

$$x = 4 + e^{-t}(\cos t + \sin t), \qquad y = e^{-t}(\cos t - \sin t)$$

for $0 \leq t \leq \frac{\pi}{2}$. The curve is rotated 2π radians about the x-axis.

b) Find the area of the surface area formed giving your answer in exact form.